Jinwon Lee

AI Platform Software & SRE Engineer

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"Build it first, then think it through."

TARGET FIT

- AI Platform Software
- MLOps / Model Serving
- Kubernetes Infrastructure
- System Software

SUMMARY

At Samsung Electronics, I work on SRE and platform engineering for generative AI services and internal LLM platforms. For about eight years, I have built and operated Kubernetes-based services, cloud infrastructure automation, CI/CD, observability, and AI evaluation/serving workflows.

My strength is taking AI applications beyond "working code" and turning them into systems that real users can rely on. I consider deployment, authentication, monitoring, incident response, and resource constraints together to build platforms where applications can run smoothly and reliably.

AI Platform Software | MLOps & Model Serving | System Software & Performance | Cloud/Kubernetes Infrastructure

CORE SKILLS

AI/MLOps	LLM evaluation pipeline, MLflow metric tracking, AI serving operations, Document LLM platform, Langfuse observability
Platform	Kubernetes, Azure, AWS, OpenShift, Docker, Helm/Helmfile, ArgoCD, GitOps, Terraform, Ansible
System	Linux runtime behavior, Python automation, high-throughput packet mirroring, distributed systems, resource-aware optimization
Reliability	Grafana, Prometheus, OpenTelemetry, Loki, Keycloak/OAuth, DevSecOps checks, incident-aware operations

EXPERIENCE

Samsung Electronics (AX Development Group)

Seoul, South Korea

SRE / AI Platform Engineer — Jan. 2026 - Present

- Architected and managed a global-scale B2C generative AI service on Microsoft Azure in collaboration with KT and Microsoft, covering network design and multi-region readiness.
- Optimized AI serving performance and service efficiency while designing security policies and infrastructure governance for production AI workloads.
- Engineered GitOps-based CI/CD pipelines using ArgoCD and Python, integrating static code analysis and container image scanning for DevSecOps.
- Led Kubernetes-based MLOps and serving operations for browser-integrated AI and Document LLM platforms, including authentication and observability frameworks with Keycloak, OAuth, and Langfuse.

Samsung Research

Seoul, South Korea

SRE / AI Platform Engineer — Jul. 2024 - Dec. 2025

- Operated a company-wide AI chatbot service on internal Kubernetes clusters, owning platform and application reliability for production use.
- Led infrastructure, platform, and application operations for a company-wide AI chatbot platform serving 30,000 employees on Kubernetes.
- Provisioned and operated development, staging, and production Kubernetes clusters using Helm and Helmfile.
- Built automated CI/CD pipelines for containerized applications with Docker, Kubernetes, GitHub Actions, and ArgoCD.
- Deployed centralized monitoring with Grafana, Prometheus, and OpenTelemetry, and built an automated LLM evaluation pipeline for Machine Translation tasks.

EXPERIENCE (CONTINUED)

Samsung Electronics (Network Dept.)
Seoul, South Korea

SRE & Software Engineer
Jan. 2021 - Jun. 2024

- Deployed containerized applications using Docker, Helm, Kubernetes Operator, and AWS services including EKS, ECS, Route 53, RDS, and IAM.
- Built and managed OpenShift, Kubernetes, and EKS environments with focus on performance, reliability, high availability, and auto-scaling.
- Developed an automated one-click Kubernetes cluster provisioning system for cloud developers.
- Developed an application that mirrors up to 130,000 packets per second in real time under constrained system resources.

TmaxOS & TmaxCloud
Gyeonggi, South Korea

Software & Platform Engineer
Jan. 2019 - Dec. 2020

- Developed CI/CD and registry operators in Kubernetes and deployed a private container image registry.
- Built a Siri-like AI assistant on TmaxOS with system-level integration, AI model inference, preprocessing, and asynchronous architecture.

EDUCATION

UNIST (Ulsan National Institute of Science and Technology)
Mar. 2017 - Feb. 2019

M.S. in Computer Science and Engineering

- Master Thesis: Convergence Aware CNN Training
- Advisors: Jiwon Seo (Hanyang University), Sam H. Noh (UNIST)

UNIST (Ulsan National Institute of Science and Technology)
Mar. 2012 - Feb. 2017

B.S. in Computer Science and Engineering; Minor in Mathematics

- Undergraduate research: distributed systems for AI model training with TensorFlow

CERTIFICATIONS & LANGUAGE

- OPIc IH (Intermediate High), Dec. 2025

PUBLICATIONS

- Alleviating the Network Bottleneck for CNN Distributed Training through Automatic Resource-Aware Layer Placement, USENIX NSDI 2019 poster.
- Accelerated Training for CNN Distributed Deep Learning through Automatic Resource-Aware Layer Placement, arXiv 2019.
- Improving Performance of Distributed TensorFlow using CNN Characteristics Exploiting Model Parallelism, ACM EuroSys 2018 poster.